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Predicting risk in energy markets: Low-frequency data still matter☆

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ABSTRACT

Are high-frequency data always needed to generate precise forecasts of risk measures in energy markets? This study attempts to shed light on this question. We study whether energy market participants can rely on low-frequency volatility estimators when interested in two market risks: volatility and expected shortfall. Using ten years of data on four of the world's most liquid energy futures contracts – the crude oil benchmarks West Texas Intermediate and Brent, as well as natural gas and heating oil futures – we provide conclusive evidence that while realized volatility models lead to much more accurate forecasts in the short term, medium- and longer-term forecasts based on daily ranges are comparable and, in some cases, even more accurate than their high-frequency counterparts that are computationally more intensive and that require costly data. Next, we present an application to predict extreme price declines - expected shortfall - with low-frequency volatility estimates. For that purpose, we propose a novel complete subset quantile regression model to predict multiple-day-ahead expected shortfall . Our back-testing results show that the new model leads to well-specified price decline forecasts, particularly when used with low-frequency volatility estimates. These results show that depending on the forecast horizon and purpose, low-frequency, publicly available, free of cost and easy to process volatility estimators still matter.

1. Introduction

Energy markets have been a central point of attention among industries, investors, regulators and academics for many decades. Understating drives of market movements is fundamental to all policy actions. Several recent studies provide example of very different approaches to such academic endeavours. For example [2] has recently employed advanced machine learning methods, while Tang et al. [3] relied on search engine data to predict oil prices. Miljkovic and Goetz [4] shows how different shocks to oil prices on the futures markets effect spot oil prices. Energy asset prices correlate with worldwide industrial production as well as macroeconomic variables (e.g., [5,6]), and energy market volatility is known to be strongly related to the stock market (e.g. [7]; and [8] offer comprehensive literature reviews on the relation between oil and equity markets). Fossil fuels are increasingly bearing stranded asset risks due to climate change concerns,[1] which adds further complexity to the volatile environment of energy assets, along with production, geopolitical and regulation aspects. For these reasons, the modeling of energy assets' volatility is an area of sustained academic endeavors.

Recent years have observed a growing body of literature in this domain, with a main focus on realized volatility (RV), a measure based on high-frequency financial market data. In general, we expect that applying RV in volatility models should lead to the most accurate volatility and market risk forecasts. However, it requires the availability of intraday time series, and such data are costly and therefore not available to all participants in the energy industry. Moreover, managing high-frequency data requires more time and computational costs. On the other hand, low-frequency daily price data are publicly available and are much simpler to work with than high-frequency data, which begs the question: 'Do we always need high-frequency data when predicting risk in energy markets?'. Our study is the first to answer this question using data from four major energy markets. We compare the predictive performance of volatility models that apply high- or low-frequency (based on daily highs and lows) volatility estimates at various forecast horizons.

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[1] Refer to Curtin et al. [1] for a review of the literature on stranded asset risk.

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Originally introduced by Andersen and Bollerslev [9], the concept of intraday data-based realized measures has been largely embraced by financial econometricians, resulting in a prolific number of estimators (e.g. [10] compare the estimation performance of 400 different realized estimators), adjustments to reflect the reality of markets (e.g., [11–13]), and the development of sophisticated model specifications (e.g., [14, 15]). Overall, realized volatility enjoys overwhelming popularity across asset classes, individual assets and applications but requires reliable and often cost-intensive time series, which may pose a challenge to practitioners.

The notion that daily ranges can be used to form volatility estimates has a long history in finance. Garman and Klass [16], Parkinson [17] and Rogers and Satchell [18] introduce range-based volatility estimators. Under ideal conditions (i.e. continuous trading, no microstructure noise or other market frictions), the range-based estimators of volatility are substantially more efficient than the traditional estimators based on daily closing prices [19]. Furthermore, volatility obtained from daily price ranges requires a small dataset that is often free of cost. Interestingly, the literature has taken a longer time to examine the properties of price ranges (e.g., [20,21]) as well as the relative performance of individual range-based estimators (e.g., [19,22,23]) in a finance context. Recent literature has introduced range-based univariate models of volatility (e.g., [24,25]) or multivariate models (e.g., [26–28]).

The literature on forecasting energy asset volatility has been abundant in the last few years with a main focus set on oil. Sévi [29], Gong and Lin [30], Liu et al. [31] and Ma et al. [32] investigate the forecasting performance of various specifications of heterogeneous AutoRegressive- (HAR-)type models for futures or spot price data on the Western Texas Intermediate (WTI) sweet crude oil. Chatziantoniou et al. [33] study the usefulness of harnessing futures data to forecast the realized volatility of the spot prices of Brent crude oil. Haugom et al. [34], Gong and Lin [35] and Lv [36] analyze the contribution of the crude oil volatility index in HAR models to forecast the realized volatility of crude oil. Ma et al. [37] investigate the predictive power of uncertainty and market sentiment indexes, macroeconomic fundamentals and technical indicators with respect to future monthly oil price volatility. Yang et al. [38] analyze the role of investor sentiment and the leverage effect for the volatility forecasting of crude oil futures in a HAR model setting. Finally, Zhang et al. [39] compare the predictive ability of forecast combination and shrinkage methods for oil price volatility. Ji and Guo [40] investigate the effects of various types of oil-related events on world oil prices, using an event study methodology and time series models. Mei et al. [41] study the effects of geopolitical risk uncertainty on oil futures price volatility within a mixed data sampling (MIDAS) modeling framework. Based on the standard forecasting as well as combination approaches, and shrinkage methods, Liang et al. [42] show that economic uncertainty indices have significant predictive power for crude oil market volatility. In contrast, fewer studies analyze other energy commodities, such as heating oil, natural gas and gasoline. Sadorsky [43] runs a horse race of a number of forecasting models using daily price returns on WTI, heating oil, unleaded gasoline, and natural gas between 1988 and 2003. Prokopczuk et al. [44] fit various HAR model specifications for WTI crude oil, gasoline (RBOB), heating oil, and natural gas using intraday data sampled from 2007 to 2012. Lyócsa and Molnár [45] study whether the predictability of volatility models for oil and natural gas can be improved by exploiting the interrelationships between these commodities for the period from 2008 to 2014.

Most of the above mentioned studies employ intraday data to generate daily volatility estimates. In contrast, range-based volatility estimation has received less attention. Ma et al. [46] focus on the impact of jump intensity on the volatility of futures in the oil futures market using HAR models for realized range-based volatility. No studies have aimed to document the differences in the performance of realized volatility and daily ranges for energy assets. If daily ranges are found to fare similarly to forecasts harnessing realized volatility series, this result will have strong practical implications, as daily ranges are less computationally and cost intensive.

Our work extends the study of Degiannakis and Livada [47] who compare daily price ranges with realized volatility estimates and show that the high-low price (open-high-low-close price) range can be more accurate than the realized volatility estimator based on five, (eight) or less, equidistant intraday points in time. These results advocate the use of daily ranges when high-frequency data are not available and serve as further motivation of our study. While Degiannakis and Livada [47] use simulations and focus on volatility estimation rather than forecasting performance, we apply a variety of forecasting techniques to real financial market data and account for market frictions.

Our work is furthermore related to Ma et al. [48] and Zhang et al. [49]. Ma et al. [48] utilize forecast combination strategies that combine low-frequency and high-frequency volatility models and find that this combination results in significantly better performance than other models and combination forecast strategies in the case of the Shanghai Stock Exchange Composite Index. Zhang et al. [49] propose a mixed-frequency approach with a model switching mechanism that switches between GARCH-class models using low-frequency data only and HAR-RV models using data for the S&P 500. In both cases, volatility forecasting is obtained by *combining* high- and low-frequency data, i.e., the best-performing approach requires the availability of intraday data. Our study is substantially different from this previous work in one crucial point, as we aim to understand to what extent the *sole* use of daily ranges is comparable to the performance of the unarguably very potent realized metrics obtained from intraday data. Using ten years of intraday data on four of the world's most liquid energy futures contracts, we compare the performance of high- and low-frequency volatility models using three classes of models: (i) the popular HAR model and its extensions, (ii) the ARFIMA-GARCH model with variance measures in the mean equation, and (iii) realized/range-GARCH models with asymmetric return distributions. Furthermore, we analyze the performance of averaging individual forecasts. This approach is motivated by Bates and Granger [50] and Timmermann [51], who advocate that a forecast generated by combining predictions from different models is suitable to average out the model choice uncertainty. Various combination methods have been proposed in the literature and have been found to conclusively improve the forecasting accuracy of single models (e.g., [52–56]). For all statistical evaluations, we use a high-frequency data-based volatility benchmark.

The results show that high-frequency data-based forecasts perform significantly better in the shorter term, while the corresponding low-frequency volatility forecasts are biased. However, for medium- and longer-term forecasts, forecasts requiring solely daily (low-frequency) data can exhibit performance that is statistically indistinguishable from that of realized volatility models. This finding is documented for all assets under consideration and for forecasting horizons longer than two weeks. A comparison of individual and combination forecasts obtained with both high- and low-frequency data confirms that combination forecasts improve the forecasting performance of individual models. These results cast doubt that the data- and computationally more intensive realized volatility measures are needed for medium- to long-term forecasting purposes.

While volatility forecasts show general market risk tendencies, stakeholders and, particularly, market participants are interested in potential future extreme price declines. For example, the production of miners and the government budgets of oil and gas exporting countries might be adversely affected during unexpected market price declines. Reliable prediction of such market declines is thus of great importance to all stakeholders in the supply chain. We therefore present an application on how to use publicly available, free of cost, low-frequency volatility estimates to predict extreme price declines in energy markets. Specifically, we use realized volatility and range-based estimators to predict the multiple-day-ahead 5% expected shortfall (ES).

While there are a few studies comparing the performance of high- and low-frequency data based Value-at-Risk forecasts (e.g., [57,58]), less attention has been devoted to multiple day-ahead ES in this context. Degiannakis and Potamia [59] compare the accuracy of models utilizing daily and intraday returns based on a sample of stock indices, commodities (industrial and precious metals), foreign exchange rates for both Value-at-Risk and ES. They establish that there is no unique superior model for all cases and intraday volatility measures do not seem to outperform lower frequency volatility proxies when the time horizon of the estimation increases. Our contribution in this context is manifold. We extend the discussion to some of the world's most liquid energy futures contracts. To that end, we present a new approach to predict ES, the complete subset quantile regression expected shortfall model (CSQR-ES). Moreover, we compare the performance of the model when applied with high-frequency realized variance or low-frequency range-based variance.

We use the back-testing strategy of Couperier and Leymarie [60] and a statistical comparison based on the Fissler and Ziegel [61] loss function used in [62] ($FZ^{(0)}$ henceforth). Our results show that across several CSQR-ES model specifications, we always find multiple models that lead to well-specified forecasts of the expected shortfall. Moreover, almost all low-frequency volatility model forecasts are well-specified and lead to lower $FZ^{(0)}$ losses as opposed to losses generated by high-frequency model forecasts. We therefore find that for the purpose of ES forecasts, low-frequency volatility estimates are useful.

The remainder of the paper is structured as follows. The next section describes the data. Section 3 presents the volatility estimators and forecasting techniques as well as a model for forecasting expected shortfall. Section 4 discusses our results, and Section 5 concludes.

2. Data

This study uses intraday transaction data sampled at the 5-min frequency for the front-month continuous series for NYMEX Western Texas Intermediate (WTI) sweet crude oil, heating oil and natural gas futures and Brent crude futures traded at the Intercontinental Exchange. The data are obtained from Refinitiv Tick History (former Thomson Reuters Tick History). Each five-minute window contains the first, highest, lowest and last transaction prices recorded during this period. All data entries are denominated in USD.

Depending on the series, our data begin in 2009 and end on 31st October 2019. The natural gas series starts on 31st May 2009, heating oil on 30th June 2009, WTI on 1st January 2009 and Brent on 31st May 2009. For natural gas, heating oil and WTI, we set a trading day from 6:00 pm on one day to 5:00 pm EST on the next day, which reflects the contract specifications for trading through Globex and ClearPort (Sunday-Friday 6:00 pm-5:00 pm). Brent futures are traded for 22 h per day, from 1:00 am to 11:00 pm GMT/BST. Our samples start at different dates, after which the number of daily quotes is similar. In this way, we ensure the comparability of volatility estimates over time.

3. Methodology

In this section, we describe our technical approach. First, we present high- and low-frequency volatility estimators, which we subsequently employ in volatility and expected shortfall models. Even though daily ranges originate from the entire observed price process of one trading day and as such might be seen as linked to high-frequency data, we opt for labeling daily price ranges 'low-frequency estimators' in order to clarify that for the constructing of daily ranges, one only requires publicly available, free-of-cost data comprising of two, up to a maximum of four data points per day and no extensive datasets of intraday price/quotation records. This approach avoids any potential confusion which could arise due to the existence of the realized range estimators of Christensen and Podolskij [63] and Martens and Dijk [64], which are based on highest and lowest prices from short intraday intervals. We use the term 'high-frequency estimators' to label realized variance estimators estimated at the 5 min frequency, while the term 'ultra-high frequency' would be reserved for realized variance estimators estimated at the frequency of seconds (e.g. [65,66]).

3.1. Volatility estimators

We study how the accuracy of two classes of estimators of the energy price variation might influence market risk assessment. The two classes of estimators are (i) high- and (ii) low-frequency volatility estimators presented in the following sections. While it is generally accepted that the former is more efficient (e.g., [11,19,21]), it is unclear whether efficiency gains translate into more accurate multi day-ahead predictions of volatility.[2] and expected shortfall.

3.1.1. High-frequency estimator

To test the usefulness of several hundred volatility estimators, Liu et al. [10] compared the accuracy of these estimators for 31 assets. Their volatility forecasting study suggests that other high-frequency volatility estimators rarely out-perform the 5 min RV. Motivated by these results, we employ the 5 min RV as our high-frequency volatility estimator. Let $P_{t,j}$ denote the log price of the asset at day $t = 1, 2, \ldots, T$ and equally spaced intraday time period $j = 1, 2, \ldots, N$. The standard high-frequency volatility estimator is the 5 min realized volatility estimator of the form

$$RV_t = \sum_{j=2}^{N} \left(P_{t,j} - P_{t,j-1}\right)^2. \tag{1}$$

3.1.2. Low-frequency estimators

After the seminal papers of Andersen et al. [11,67], the research has transitioned towards the adoption of high-frequency volatility estimators. As a consequence of the immense popularity of the GARCH volatility models, the most popular low-frequency estimator appears to be the squared daily return; however, more efficient range-based estimators exist (e.g., [21,68,69]). Given the log of the opening (O_t), high (H_t), low (L_t), and closing (C_t) prices at day t, we scale variables to the opening price as $h_t = H_t - O_t$, $l_t = L_t - O_t$, $c_t = C_t - O_t$, after which the range-based estimator of Parkinson [17] is

$$PK_t = \frac{\left(h_t - l_t\right)^2}{4ln2}, \tag{2}$$

Rogers and Satchell [18] estimator

$$RS_t = h_t(h_t - c_t) + l_t(l_t - c_t), \tag{3}$$

and the Garman and Klass [16] estimator

$$GK_t = 0.511(h_t - l_t)^2 - 0.019\left[c_t(h_t + l_t) - 2l_t h_t\right] - 0.383c_t^2. \tag{4}$$

Instead of considering each of the estimators individually, we use a composite estimator as the simple average across the three estimators,

$$RB_t = \frac{PK_t + RS_t + GK_t}{3}. \tag{5}$$

The motivation for that is as follows. The Rogers and Satchell [18] estimator is a drift independent estimator, while the Garman and Klass [16] and Parkinson [17] estimators assume a zero drift price path. As in practice the true data-generating process is never known, by taking the average, we mitigate the estimator choice uncertainty (see [70], for the same idea). Motivated by the relative efficiency of the daily price ranges, Christensen and Podolskij [63] and Martens and Dijk [64] propose a method based on price ranges over intraday intervals. This approach is not the focus of this study, as realized ranges – similar to RV_t – require intraday data series.

[2] For day-ahead setting with equity indices see [48].

3.1.3. Filtering raw data and estimators

Data from the Refinitiv Tick History database are considered to be of high quality. However, we found several occasions where the daily high (low) was more than 50% above (below) the opening price. In these instances, we used the previous day's range-based estimator.

Both realized and range-based estimators were still subject to a few extremes. Unfortunately, price variation during such extreme days is driven mostly by events that cannot be anticipated by time-series models that rely on historical data. On the other hand, such days also generate extreme forecast errors that can have a large influence on the evaluation of forecasting models.

Motivated by these arguments, we run a filtering algorithm in the following way: (i) using a rolling window of 1000 observations (which corresponds to our estimation window), we created a 99.95% percentile series; (ii) if the following observations are larger than the 99.95% percentile, the value of the estimator is winsorized at the 99.95% percentile. In this way, our algorithm picks only the most extreme observations, and because it is forward looking, it can be employed in a real-world scenario. This algorithm is applied to all range-based estimators, realized variance, and other realized measures employed in the volatility models, e.g., semi-volatilities.

3.2. Volatility models

We use the high- (RV_t) or low-frequency (RB_t) volatility estimators within the three classes of volatility models that are described in the following three subsections.

3.2.1. HAR class volatility models

Denote the daily, weekly, monthly and quarterly (non-overlapping) realized volatility components as $RV_t^D = RV_t$, $RV_t^W = 4^{-1}\sum_{i=t-1}^{t-4} RV_i$, $RV_t^M = 17^{-1}\sum_{i=t-5}^{t-21} RV_i$ and $RV_t^Q = 44^{-1}\sum_{i=t-22}^{t-65} RV_i$. The h day-ahead average volatility is $RV_{t+h} = h^{-1}\sum_{k=1}^{h} RV_{t+k}$, where $h = 1, 2, \ldots, 66$. The popular heterogeneous autoregressive model of Corsi [71] is of the form

$$RV_{t+h} = \beta_{h,0} + \beta_{h,1} RV_t^D + \beta_{h,2} RV_t^W + \beta_{h,3} RV_t^M + \beta_{h,4} RV_t^Q + u_{h,t+h}. \quad (6)$$

The model in Eq. (6) is denoted either as the *RV-HAR* model if populated with high-frequency volatility estimators or as *RB-HAR* if the range-based composite estimator is used. Specifically, the range-based daily, weekly, monthly and quarterly components are $RB_t^D = RB_t$, $RB_t^W = 4^{-1}\sum_{i=t-1}^{t-4} RB_i$, $RB_t^M = 17^{-1}\sum_{i=t-5}^{t-21} RB_i$ and $RB_t^Q = 44^{-1}\sum_{i=t-22}^{t-65} RB_i$. The standard HAR model consists of the monthly volatility component at most. However, as our maximum forecast horizon is set to $H = 66$ in this study, corresponding to 3 trading months, we also include the corresponding volatility component on the right side of the equation.

The common argument for separate volatility components is based on the heterogeneous market hypothesis that states that investors with different investment horizons cause different levels and features of market volatility [72]. An alternative interpretation might be that volatility estimators are very noisy estimators, and the averaging introduced in volatility components helps to reduce this estimation uncertainty. Consequently, because volatility is persistent, a more precise estimation of past volatility might lead to more precise forecasts. One of the attractive features of the HAR model is that it tends to capture volatility persistence and long-term memory well, while it is easy to estimate and modify.

Additionally, we use two modifications of the HAR model. Let

$$RV_t^{(-)} = \sum_{j=2}^{N} \left(P_{t,j} - P_{t,j-1}\right)^2 \times I\left[(P_{t,j} - P_{t,j-1}) < 0\right] \quad (7)$$

denote negative semi-volatility, where $I[.]$ is an indicator function that returns 1 if the return is negative and 0 otherwise. The *RV-HAR-NV* model of Patton and Sheppard [73] can be specified as

$$RV_{t+h} = \beta_{h,0} + \beta_{h,1} RV_t^D + \beta_{h,2} RV_t^{D(-)} + \beta_{h,3} RV_t^W + \beta_{h,4} RV_t^M + \beta_{h,5} RV_t^Q + u_{h,t+h}. \quad (8)$$

Our specification is different from the model introduced in [73] (see their Eq. (16)), but the explanatory power of our specifications is the same as that of their specifications. The inclusion of the semi-variance component in the HAR model is motivated by the asymmetric volatility effect by which volatility tends to be larger when the market declines. The semi-volatility component is calculated from intraday prices. For this reason, a low-frequency counterpart to that model is not possible. We therefore use the following specification that we denote *RB-HAR-NV*,

$$RB_{t+h} = \beta_{h,0} + RB_t^D \times \left(\beta_{h,1} + \beta_{h,2} \times I\left[(P_{t,N} - P_{t,1}) < 0\right]\right) + \beta_{h,3} RB_t^W + \beta_{h,4} RB_t^M + \beta_{h,5} RB_t^Q + u_{h,t+h}. \quad (9)$$

A $\beta_{h,2} > 0$ coefficient implies that if the market declines, the next day's volatility has a tendency to increase more as opposed to a day when the market rises. Similar ideas were also explored in [74] or more recently in [75],

$$RV_{t+h} = \beta_{h,0} + \beta_{h,1} RV_t^D + \left|P_{t,N} - P_{t,1}\right| \times \left(\beta_{h,2} + \beta_{h,3} \times I\left[(P_{t,N} - P_{t,1}) < 0\right]\right) + \beta_{h,4} RV_t^W + \beta_{h,5} RV_t^M + \beta_{h,6} RV_t^Q + u_{h,t+h}. \quad (10)$$

Here, the asymmetric effect is controlled by the $\beta_{h,3}$ coefficient. The model is denoted as *RV-HAR-ASY* if populated with realized volatility and as *RB-HAR-ASY* if populated with the low-frequency range-based composite estimator. As noted previously, large unanticipated values of the volatility result in heteroscedastic residuals that are tied to the level of the volatility [73]. Consequently, we estimate parameters of HAR models by minimizing the weighted least squares of errors, where the weights are given by (standardized) RV_{t+h}^{-1} or RB_{t+h}^{-1}, i.e., the higher the volatility at day $t + h$, the lower the weight of that observation. Alternative estimation approaches are discussed in [76].

3.2.2. ARFIMA-GARCH class volatility model

As volatility series tend to display a long memory, Andersen et al. [67] suggested modeling realized volatility within an AutoRegressive Fractionally Integrated Moving Average model of volatility that we complement with Generalized AutoRegressive Conditionally Heteroscedastic errors (ARFIMA-GARCH henceforth), (e.g., [7,77,78]). Let L denote the lag operator and $d \in (0, 1)$ denote the fractional integration parameter. The mean equation is specified as

$$RV_t = \mu_0 + u_t; \quad (1 - \phi L)(1 - L)^d u_t = (1 - \theta L)\epsilon_t, \quad (11)$$

where ϕ and θ are autoregressive and moving average coefficients, while the error term, $\epsilon_t = \sigma_t \eta_t$, can be decomposed into an *iid* random variable $\eta_t \sim iid(0, 1)$ and a ... latent volatility σ_t that follows the standard GARCH(1, 1) volatility process of Bollerslev [79],

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \quad (12)$$

The η_t is usually assumed to follow a normal distribution and the stationarity conditions require the parameters to satisfy $\alpha + \beta < 1$. As the volatility of volatility tends to be skewed to the right [80], a more flexible distribution is needed, and we use Johnson's SU distribution [81,82], which is flexible enough to incorporate asymmetries and heavy tails. The resulting model is denoted as the *RV-ARFIMA-GARCH* model if high-frequency realized volatility is modeled in the mean equation and as *RB-ARFIMA-GARCH* if the low-frequency range-based composite estimator is used. Multiple-day-ahead forecasts are computed recursively.

3.2.3. Measurement equation GARCH volatility model

Finally, we rely on the new generation of GARCH models introduced by Hansen et al. [83] that were designed to harness the information content of realized volatility measures and allow for asymmetric shocks, while also relaxing parameter constraints by modeling

the $ln(\sigma_t^2)$ instead of σ_t^2. The mean equation is an ARMA process of open-to-close returns

$$R_t^I = \left(P_{t,N} - P_{t,1}\right) = \mu_0 + u_t; \quad (1-\phi L)u_t = (1-\theta L)\epsilon_t, \tag{13}$$

where as before, $\epsilon_t = \sigma_t \eta_t$ with η_t behaving according to the Johnson's Su distribution, σ_t now given by

$$ln(\sigma_t^2) = \omega + \alpha RV_{t-1} + \beta ln(\sigma_{t-1}^2) \tag{14}$$

and the new measurement equation is

$$RV_t = \xi + \delta ln(\sigma_t^2) + \upsilon_1 \eta_t + \upsilon_2(\eta_t^2 - 1) + u_t; \; u_t \sim N(0, \lambda), \tag{15}$$

with the term $\upsilon_1 \eta_t + \upsilon_2(\eta_t^2 - 1)$ accounting for asymmetric shocks. Multiple-day-ahead forecasts are created using recursive methods. If the measurement equation is populated by the high-frequency realized volatility estimator, the model is denoted as *RV-GARCH*, and in the case of the low-frequency range-based composite estimator, it is denoted as the *RB-GARCH* model.

3.3. Combinations of volatility forecasts

In the previous sections, we describe five high- and five low-frequency volatility models. In this section, we describe how the outcomes of individual forecasting models are combined into a single *combination forecast*. The use of combination forecasts is motivated by the desire to study whether we can harness information from different low-frequency volatility models into a single forecast that might outperform individualforecasts. As argued by Elliott and Timmermann [84], the true data-generating process is unknown, and equally importantly, it is unlikely that we can use one individual model to accurately approximate the true data-generating process. Combining multiple individual forecasts into a single forecast is a straightforward strategy to address this model choice uncertainty [51]. Furthermore, Ma et al. [48] explore how combining individual volatility forecasts within and between low- and high-frequency volatility models improves forecasts, and they present empirical evidence that for two broad stock market indices, forecast combinations are indeed useful.

We use three combination forecasts for low-frequency volatility models, three for high-frequency volatility models and three for low- and high-frequency volatility models (e.g., 9 combination forecasts). We denote ordered forecasts of $m = 1, 2, \ldots, M$ volatility models as $\widehat{RV}_{t+h}^{[m]}$ and define the first combination forecast as

$$\widehat{RV}_{t+h}^{Trim(p)} = \frac{1}{(M-2p)} \sum_{m=p+1}^{M-p} \widehat{RV}_{t+h}^{[m]}, \tag{16}$$

which is a trimmed mean, where the top p and lowest p forecasts are skipped. This combination ignores how the models have performed in the past; instead, it simply assumes that predictions with the largest deviations from the rest of the models are less accurate.

To account for the historical accuracy of individual models, we use a weighted average across forecasting models. Let L_{j+h}^m denote a loss (to be defined later) for model m, with a property that lower value of the loss means a more accurate prediction, $\delta \in (0, 1)$ is the discount factor and S the size of the calibration sample. The non-normalized weights are

$$w_{t+h}^m = \frac{1}{S} \sum_{j=t-S}^{t-1} \delta^{t-j} L_{j+h}^m. \tag{17}$$

Here, the most recent loss receives a higher weight, e.g., if a model has recently been performing poorly, its weight will be smaller. Normalized weights that sum to 1 are

$$w_{t+h}^{*m} = \frac{(w_{t+h}^m)^{-1}}{\sum_{j=1}^{M}(w_{t+h}^j)^{-1}}, \tag{18}$$

and the weighted average is

$$\widehat{RV}_{t+h}^{Disc(\delta)} = \sum_{m=1}^{M} \widehat{RV}_{t+h}^m w_{t+h}^{*m}. \tag{19}$$

A $\delta = 1.00$ would imply no weighting across observations in time, and the lower the δ is, the higher the weight of recent losses. We set the discount factor to $\delta = 0.90$. The size of the calibration sample is set to $S = 200$, which means that the past 200 out-of-sample observations are used to assess the accuracy of the volatility models.

The third combination forecast is defined as

$$\widehat{RV}_{t+h}^{Top(1)} = \sum_{j=1}^{M} \widehat{RV}_{t+h}^j \times I(\max_m[w_{t+h}^{*,m}] = w_{t+h}^{*,j}), \tag{20}$$

where the preferred forecast is the one with the lowest average loss over the calibration sample. If more than one model satisfies this condition, a preferred model is chosen randomly.

3.4. Volatility forecasting procedure

To predict volatility, we use a rolling window approach with 1000 observations. Each individual model is re-estimated after each trading day. As combination forecasts require a calibration sample, our first forecast is generated after the initial 200 observations, i.e., the first forecast is for the 1201th trading day and onwards.

3.5. Volatility forecast evaluation

To evaluate volatility forecasts from low- and high-frequency volatility models, we use the 5-min realized volatility as the proxy for the true unobserved volatility. In this way, low-frequency models are at a disadvantage because they predict the range-based composite estimator, while high-frequency models predict the realized volatility directly. We believe that in an empirical setting this is the only sensible way to make direct volatility comparisons to test whether low-frequency volatility models are useful (for a similar approach see [48]). A non-empirical approach could be based on simulated data. Volatility forecasts are evaluated along two lines: a descriptive evaluation using a loss function and a statistical test that compares forecasts across multiple models.

Patton [85] finds that only two loss functions lead to consistent model rankings in the presence of a noisy volatility proxy, namely, the mean square error

$$L_{MSE,t+h}^m = (RV_{t+h} - \widehat{RV}_{t+h})^2, \tag{21}$$

and the QLIKE [86],

$$L_{QLIKE,t+h}^m = \frac{RV_{t+h}}{\widehat{RV}_{t+h}} - ln\left(\frac{RV_{t+h}}{\widehat{RV}_{t+h}}\right) - 1. \tag{22}$$

In this manuscript, we present results from the QLIKE loss function only, while results for the MSE lead to the same conclusions and are therefore available upon request. Before evaluating forecasts, we substitute all negative forecasts with the lowest observed realized variance (or range-based estimator) up to that point. We also winsorize all forecasts above the historical 99.9% percentile. In the tables and figures, we report the average values of QLIKE losses.

Statistical evaluation follows the model confidence set approach of Hansen et al. [87], which is used to sequentially evaluate competing volatility models until models with equal predictive accuracy remain. The test is very useful in our context in two ways. First, the sequential testing framework leads to a superior set of models (forecasts), which will include only high-frequency volatility models if low-frequency volatility models cannot match the forecasting accuracy of the former. Second, as we are comparing 19 models (5 high-frequency models, 5 low-frequency models, 3 combination forecasts from high-frequency models, 3 combination forecasts from low-frequency models and 3

combination forecasts from both low- and high-frequency models), our testing might be subject to data-snooping bias, which is addressed by the sequential procedure of Hansen et al. [87]. More specifically, we use the T_{max} statistic with loss function variance estimation and critical value derivation based on the stationary bootstrap with random block lengths (derived using the method of Politis and Romano [88] and Patton et al. [89]) using 5000 samples and $\alpha = 0.15$.

3.6. Forecasting expected shortfall: The CSQR-ES model

Expected shortfall (ES henceforth) represents the expected return, conditional on returns falling below some pre specified quantile τ, i.e., below the Value-at-Risk (VaR henceforth). More specifically,

$$\begin{aligned} VaR_t(\tau) &= Q_{R_t}(\tau, I_{t-1}) \\ ES_t(\tau) &= E\left[R_t | R_t \le VaR_t(\tau), I_{t-1}\right], \end{aligned} \tag{23}$$

where I_{t-1} is the available information at time $t-1$ and $Q_{R_t}(.)$ is a quantile function of daily, close-to-close returns R_t. The ES is a risk measure applied by investors, businesses and regulators to manage financial risks from market positions. After the financial crisis in 2008, many felt that VaR severely underestimated potential losses [90]. The VaR can be understood as a lower bound to ES. Therefore, the Basel Committee on Banking and Supervision has promoted ES as a key measure to use instead of the VaR measure to calculate the capital requirements of banks.

Standard models to estimate ES are based on AR-GARCH models (e.g., [91]). In this study, it is not our goal to find the most accurate model. Instead, our goal is to compare the accuracy of low- and high-frequency volatility estimators for the purpose of estimating the expected shortfall of a traded energy asset.[3] For that purpose, we use an approach to estimate the expected shortfall that is based on estimating and aggregating VaR using quantile regressions. This approach allows us to use either low- or high-frequency volatility estimates and therefore compares the usefulness of high- and low-frequency estimators in predicting ES straightforwardly.

The estimation and testing strategy is based on the formulation of ES as a function of VaRs and its approximation,

$$\begin{aligned} ES_t(\tau) &= \frac{1}{\tau}\int_0^{\tau} VaR_t(u)du, \\ ES_t(\tau) &\approx \frac{1}{p}\sum_{j=1}^{p} VaR_t(u_j), \end{aligned} \tag{24}$$

where u_j corresponds to $j = 1, 2, \dots, p$ quantile levels, $u_j = 1 - \left[(1-\tau) + (j-1)(1-(1-\tau))p^{-1}\right]$. To estimate ES_t, we need to estimate p numbers of $VaR(u_j)$. Our strategy to backtest ES follows [60] in that we will consider ES to be valid if the sequence of $VaR_t(u_j)$ itself is valid.

In the following section, we first describe the VaR predictions, followed by the backtesting of ES.

3.6.1. CSQR-VaR: Complete subset quantile regression model to predict Value-at-Risk

We model daily returns R_{t+h}, for $h = 1, 2, \dots, 66$ as a linear function of a set of K variables, $x_t = (x_{1,t}, \dots, x_{1,K})'$, $z_t = (1, x_t')$. The u_jth conditional quantile of the asset return is $z_t'\beta_h(u_j)$, $P(R_{t+h} \le z_t'\beta_h(u_j)|x_t) = u_j$. The $\beta_h(u_j)$ is a vector of coefficients that likely varies with u_j, implying nonlinear relationships between returns and the explanatory variables. To estimate the coefficients, we use the asymmetric loss function:

$$\rho(u_j, v) = v(u_j - I(v < 0)) = 2^{-1}\left[|v| + (2u_j - 1)v\right], \tag{25}$$

that is minimized,

$$\hat{\beta}(u_j) = arg\ min_{\beta(u_j)} \sum_{t=1}^{T} \rho(R_{t+h} - z_t'\beta_h(u_j)). \tag{26}$$

Elliott et al. [92] proposes a complete subset linear regression as a method to combine forecasts from a complete subset of k-variate regressions, where $k \le K$. The idea is to estimate every possible specification with $k \le K$ regressors and combine the resulting forecasts into a single forecast. The combination can be employed using simple or weighted averages. Meligkotsidou et al. [93,94] recently incorporated a complete subset linear regression into a quantile regression framework, creating the so-called complete subset quantile regression (CSQR henceforth). More specifically, Meligkotsidou et al. [93] uses the CSQR to predict quantiles of the equity premium, and in Meligkotsidou et al. [94], a similar approach is employed to predict month-ahead market volatility.

We use the CSQR to predict the quantiles necessary to estimate the $\tau = 5\%$ ES. To estimate the ES, we predict $p = 5$ VaRs, which leads to the requirement to predict the following five quantiles of returns ($5\%, 4\%, 3\%, 2\%, 1\%$ using $1 - \left[(1-\tau) + (j-1)(1-(1-\tau))p^{-1}\right], j = 1, 2, \dots, p$). Choosing lower values of τ or higher values of p leads to the requirement of estimating even lower quantiles, which comes at the expense of lower estimation precision. For example, with $p = 6$, we need to estimate 5.00%, 4.17%, 3.33%, 2.50%, 1.67%, 0.83%.

To compare the accuracy of high- and low-frequency ES predictions, we estimate complete subsets of regressions generated by (i) lagged close-to-close returns R_t, (ii) lagged intraday returns multiplied by an indicator function $\left|P_{t,N} - P_{t,1}\right| \times I\left[(P_{t,N} - P_{t,1}) < 0\right]$ and the corresponding averages over the previous five, twenty-two and sixty-six days. To compare high- and low-frequency ES models, we use either

- (iii) high-frequency volatility estimators RV_t^D, and (iv) the negative semi-volatility $RV_t^{D(-)}$, both also with corresponding averages over the previous five, twenty-two and sixty-six days

or

- (iii) low-frequency volatility estimator RB_t^D, and the (iv) interaction term $RB_t^D \times I\left[P_{t,N} - P_{t,1} < 0\right]$, both also with corresponding averages over the previous five, twenty-two and sixty-six days.

In both instances, we have $K = 13$ potential regressors. Predictions from the first set of potential explanatory variables also utilize high-frequency data, while the second set consists only of low-frequency data. A similar approach to estimate VaR_t using a quantile regression through estimates of market volatility was employed by Haugom et al. [95]. Next, we use the following complexity parameters $k = 1, 2, 3$. Given $K = 13$ potential regressors, this leads to $K!/((K-k)!k!) = 13$ quantile regressions with a $k = 1$ parameter, 78 models with $k = 2$ parameters and 286 models with $k = 3$ parameters, or $M = \sum_{k=1}^{3} K!/((K-k)!k!) = 377$ models together. Predictions of return R_t for a given quantile (u_j), forecast horizon $h = 1, 2, \dots, 66$ and model $m = 1, 2, \dots, M$ are denoted as $\hat{R}_t^{h,m}(u_j)$.

To arrive at a single prediction of a given quantile, we use four combination methods. The first is the simple average across all 377 models

$$VaR_{t,A}^{h,H}(u_j) = \frac{1}{M}\sum_{m=1}^{M} \hat{R}_t^{h,m}(u_j). \tag{27}$$

The index H denotes VaR based on high-frequency data. The $VaR_{t,A}^{h,L}(u_j)$ is the analogous version of low-frequency data forecasting.

Next, we use the (Discounted Asymmetric Loss Forecast Error) DALFE method of Meligkotsidou et al. [96] to evaluate the fit of each of the m prediction models (for each quantile separately). More specifically, we use the following weight assigned to each model:

$$w_t^{h,m}(u_j, \psi) = \sum_{j=t-s}^{t-1} \rho(u_j, R_j - \hat{R}_j^{h,m}(u_j))\psi^{t-j}, \tag{28}$$

[3] A study that compares the accuracy of predicted ES from multiple models, including the CSQR-ES model, is left for future research.

where ψ is a discount factor that gives recent losses higher weight. The weights are re-scaled to sum to one,

$$w_t^{h,m,*}(u_j,\psi) = \frac{\left[w_t^{h,m}(u_j,\psi)\right]^{-1}}{\sum_{j=1}^{M}\left[w_t^{h,j}(u_j,\psi)\right]^{-1}}. \tag{29}$$

Using weights $w_t^{h,m,*}(u_j,\psi)$, the VaR predictions are given as

$$VaR_{t,\psi}^{h,H}(u_j) = \sum_{m=1}^{M} w_t^{h,m,*}(u_j,\psi)\hat{R}_t^{h,m}(u_j). \tag{30}$$

3.6.2. Expected shortfall estimation

To estimate VaR, we use a rolling window forecasting framework with an estimation window of 1000 observations. To calculate the weights defined in the previous section, we need out-of-sample VaR forecasts. For that purpose, we use additional 250 out-of-sample VaR forecasts (the hold-out sample). Therefore, our first ES prediction is for the 1251th observation. The next step is created by rolling the estimation window by one day and repeating the estimation procedure.

Using the estimated VaRs from the CSQR and approximation given in Eq. (24), we estimate ES, which we refer to as the CSQR-ES model prediction. Based on the combination approach used and whether we employ high- or low-frequency volatility estimates, the resulting ES predictions are denoted as $ES_A^{h,H}$, $ES_{\psi=0.5}^{h,H}$, $ES_{\psi=0.9}^{h,H}$, $ES_{\psi=1.0}^{h,H}$ for high-frequency volatility estimates, and for low-frequency volatility estimates, ES predictions are denoted as $ES_A^{h,L}$, $ES_{\psi=0.5}^{h,L}$, $ES_{\psi=0.9}^{h,L}$, $ES_{\psi=1.0}^{h,L}$ for forecast horizons $h = 1, 2, \ldots, 66$.

3.7. Backtesting expected shortfall

Our backtesting strategy is based on estimating the following multiquantile regression model:

$$\hat{\gamma}^h = arg\ min_{\gamma^h} T^{-1}\sum_{t=1}^{T}\left(\sum_{j=1}^{p}\rho(u_j, R_t - \gamma_0^h(u_j) - \gamma_1^h(u_j)VaR_t^h(u_j))\right). \tag{31}$$

The dependent variable is the realized daily return, and the $VaR_t^h(u_j)$'s correspond to out-of-sample predicted quantiles. The multiquantile approach of Couperier and Leymarie [60] is the generalization of Gaglianone et al. [97], who introduced VaR as a regressor into a single quantile regression model when evaluating single VaR forecasts.

The coefficients of interest are $\gamma_0^h(u_j)$ and $\gamma_1^h(u_j)$. If $\gamma_0^h(u_j)$ are not 0, they indicate a systematic under- or over-estimation of the quantiles and, hence, the ES. If $\gamma_1^h(u_j)$ is not equal to 1, it indicates a correlation between forecast errors and quantile series. Instead of testing each coefficient separately, Couperier and Leymarie [60] devises several tests. We use the Joint Test 2, JT_2 where H_0:

$$\sum_{j=1}^{p}\gamma_0^h(u_j) = 0$$

and

$$\sum_{j=1}^{p}\gamma_1^h(u_j) = p.$$

We report test statistics and p-values derived using a stationary bootstraps algorithm where block lengths are drawn from a geometric distribution with a block length given as in [89] and having 1000 replications.

3.8. Statistical comparison of CSQR-ES models

We also perform an explicit statistical comparison of CSQR-ES from all our models. We follow the work of Fissler and Ziegel [61] and particularly [62], who proposes a loss function to evaluate the ability of a model to predict VaR and the ES. Given that $ES_t^h(\tau) \le VaR_t^h(\tau) < 0$ and $L_t = I(R_t < VaR_t^h(\tau))$ as an indicator function, Patton et al. [62] proposes the loss function

$$FZ_t^{h,(0)} = \frac{L_t(R_t - VaR_t^h(\tau))}{\tau ES_t^h(\tau)} + \frac{VaR_t^h(\tau)}{ES_t^h(\tau)} + log(-ES_t^h(\tau)) - 1. \tag{32}$$

The lower the average value of $FZ_t^{h,(0)}$ is, the better the model. The statistical comparison across all eight ES predictions and corresponding VaR at given forecast horizon is performed via the MCS test of Hansen et al. [87], as described in Section 3.5 using $\alpha = 0.15$ (for a recent application of the MCS test in the context of ES, see [98]).

4. Results

4.1. Preliminary data analysis

Table 1 reports summary statistics for the daily realized volatilities (RV_t, Eq. (1)), the daily composite range-based volatility estimate (RB_t, Eq. (5)) and the daily close-to-close (R_t) and open-to-close (R_t^I) returns for the four energy futures under consideration. During weekdays trading is almost continuous and therefore open-to-close returns (to estimate GARCH models) and close-to-close returns (to estimate expected shortfall) show slightly different properties mainly due to weekends (see Table 1). The volatility measures for all assets show positive skewness and large kurtosis and are in line with documented stylized facts of financial market volatility. Furthermore, the volatility series show a pronounced persistence that declines slowly with time. While the range-based (low-frequency) daily volatility also shows long persistence, it is markedly smaller.

Given the more discrete nature of the range-based estimator, we expect the realized variance to be larger than the range-based estimator. Descriptive measures in Table 1 show that the median of the range-based estimates is indeed always smaller. The same holds for the estimates' averages for all assets except heating oil. A closer inspection revealed that without using our filtering procedure (see Section 3.1.3), range-based estimators would be much higher on average than the realized measures and would show less persistence. It is therefore advised to carefully inspect the range-based volatility series before using it.

Range-based estimates show higher positive skewness and higher kurtosis, while consequently, extremes are much larger. As theory suggests, range-based estimators are more noisy, and this is visible in Table 1 and in Fig. 1, which illustrates the course of daily volatility measures over time. The range-based estimators in Fig. 1 show much greater fluctuations. Still, the correlations between realized and range-based estimates range from 0.76 to 0.86 across assets (Panel C of Table 1). We therefore infer that while the high- and low-frequency volatility series generally exhibit a joint course, they have, to some extent, different dynamics.

These preliminary results are important, as they suggest that for short-term forecasts, low-frequency volatility models are likely to be biased (smaller average estimates) and should perform poorly (a higher standard deviation of range-based measures). This supports our motivation for using multiple day-ahead forecasting horizons, as it will likely take multiple days for low-frequency volatility forecasts to stabilize.

Among the four energy assets, natural gas appears to have the highest daily variation in terms of both high- and low-frequency volatility estimates, which is a well-known fact for this energy asset. This phenomenon is, again, related to a lower persistence of the volatility time series. In contrast, heating oil has the lowest average daily price variation. The two international oil benchmarks, WTI and Brent, show different volatility dynamics, with WTI exhibiting higher average daily volatility, which may be due to WTI contracts being traded an hour longer per trading day than Brent futures, along with other factors.

Table 1
Descriptive statistics of variance estimators and daily returns.

	WTI				Brent				Natgas				Heating oil			
	RV_t	RB_t	R_t	R_t^I	RV_t	RB_t	R_t	R_t	RV_t	RB_t	R_t	R_t^I	RV_t	RB_t	R_t	R_t^I
Panel A: Descriptive measures of variables																
Mean	1,004.79	1,003.20	0.03	−0.02	832.08	735.52	0.01	0.02	1,584.52	1,439.17	0.03	−0.05	649.63	652.78	0.02	−0.01
SD	1,301.00	1,458.16	2.16	2.09	837.22	946.82	1.87	1.81	1,559.67	1,770.30	2.92	2.60	609.62	795.97	1.78	1.65
Skewness	4.62	5.29	0.32	0.12	2.86	3.80	0.27	0.16	3.70	4.14	1.04	0.54	2.97	3.74	−0.09	0.08
Kurtosis	34.98	50.05	7.07	6.27	12.80	22.16	6.56	5.91	23.92	28.03	11.86	7.11	14.42	21.55	9.67	5.37
Minimum	18.30	10.57	−12.17	−12.15	45.54	18.01	−8.38	−8.10	119.81	33.40	−18.69	−15.99	36.72	12.55	−17.58	−8.95
5th percentile	181.63	115.40	−3.47	−3.35	167.41	84.28	−2.92	−2.86	387.16	219.87	−4.09	−3.97	148.29	88.62	−2.76	−2.66
Median	623.48	564.90	0.07	0.02	579.35	439.82	0.02	0.06	1,135.33	898.11	−0.08	−0.15	466.51	414.75	0.02	−0.02
95th percentile	3,207.64	3,331.09	3.29	3.15	2450.33	2357.89	2.85	2.77	4,124.12	4,236.14	4.56	3.95	1750.32	2025.96	2.84	2.63
Maximum	16,639.17	24,526.19	13.84	12.77	5253.99	8262.26	12.90	10.68	17,353.41	21,948.30	30.68	18.54	4684.25	6535.09	11.80	9.46
Panel B: Autocorrelation of variables																
$\rho(1)$	0.81	0.57	−0.03	−0.03	0.78	0.59	−0.06	−0.05	0.61	0.48	−0.08	−0.09	0.71	0.48	−0.02	−0.01
$\rho(5)$	0.72	0.50	0.00	0.00	0.65	0.37	0.02	0.02	0.51	0.38	−0.06	−0.06	0.59	0.31	0.02	0.01
$\rho(22)$	0.49	0.33	0.00	0.00	0.48	0.29	0.00	0.00	0.29	0.21	0.02	−0.01	0.46	0.26	0.00	−0.03
$\rho(66)$	0.16	0.12	−0.03	−0.04	0.24	0.14	−0.01	−0.02	0.04	0.05	−0.01	0.00	0.25	0.15	−0.03	−0.02
Panel C: Correlation between realized and range-based variance measures																
	0.84				0.86				0.86				0.76			

Note: Close-to-close returns (R_t) and open-to-close returns (R_t^I) are multiplied by 100, RV_t and RB_t denote the high- and low-frequency (range-based) volatility estimators, SD denotes standard deviation, $\rho(.)$ is the autocorrelation coefficient of the given order.

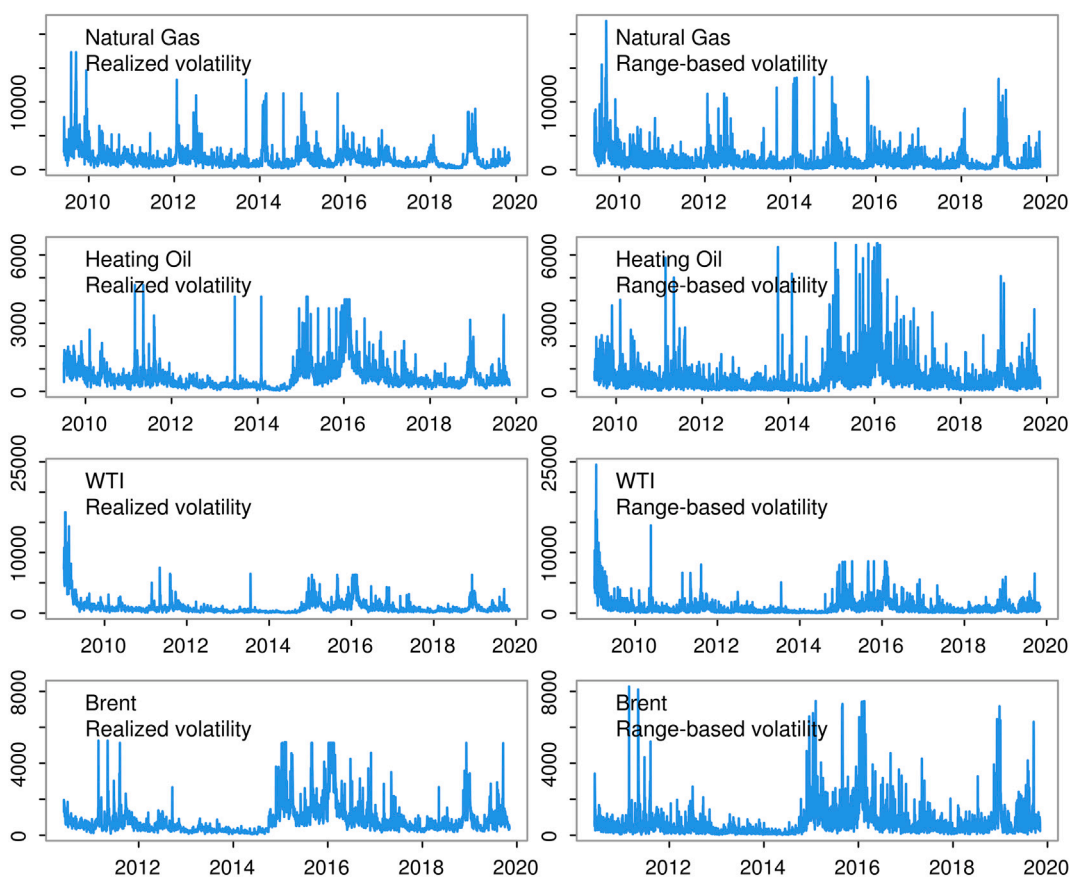

Fig. 1. Realized and range-based volatility. *Note:* The left side of corresponds to the high-frequency volatility and the right-side to low-frequency (range-based) volatility series. The comparison shows more variable behavior of the low-frequency volatility series.

4.2. Volatility forecasting models

The out-of-sample performance of individual and combination models over forecasting horizons of one, five, twenty-two and sixty-six trading days ahead is reported in Tables 2–5. The accuracy of volatility forecasts is measured against a high-frequency data-based realized volatility benchmark. The tables report average values of QLIKE, with lower values reflecting more precise volatility forecasts.

Among the individual forecasting models for realized volatility, there is no clear winner that consistently outperforms the other models across forecasting horizons and assets. Interestingly, for low-frequency volatility, the RB-GARCH model appears to perform best in almost all cases. At the same time, RB-HAR models perform poorly and are not recommended if high-frequency volatility models are not available. A possible explanation might be related to the noisiness of the range-based estimator, where instead, the latent volatility RB-GARCH

Table 2
Forecast accuracy for day-head forecasts (QLIKE).

	WTI	Brent	Natgas	HO
Panel A: High-frequency volatility models				
Standard HAR with RV	0.055	0.028	0.092	0.055
RV-HAR-NV	0.053	0.028	0.085	0.055
RV-HAR-ASY	0.058	0.03	0.091	0.057
RV-ARFIMA-GARCH	0.044[a]	0.037	0.065[a]	0.041[a]
RV-GARCH	0.087	0.038	0.096	0.071
Panel B: Low-frequency volatility models				
Standard HAR with RB	0.184	0.229	0.299	0.243
RB-HAR-NV	0.183	0.229	0.293	0.242
RB-HAR-ASY	0.19	0.231	0.295	0.243
RB-ARFIMA-GARCH	0.102	0.074	0.114	0.135
RB-GARCH	0.112	0.075	0.11	0.101
Panel C: Combining high-frequency models				
Trimmed mean	0.043[a]	0.027[a]	0.069[a]	0.045
Discounted weighted mean	0.048[a]	0.029	0.066[a]	0.04[a]
Average across top three models	0.04[a]	0.026[a]	0.063[a]	0.04[a]
Panel D: Combining low-frequency models				
Trimmed mean	0.104	0.12	0.174	0.143
Discounted weighted mean	0.098	0.068	0.089	0.089
Average across top three models	0.075	0.064	0.083	0.073
Panel E: Combining high- and low-frequency models				
Trimmed mean	0.056	0.041	0.088	0.062
Discounted weighted mean	0.048[a]	0.029	0.066[a]	0.041[a]
Average across top three models	0.047	0.03	0.067[a]	0.046

Note: The values in the table are averages of the QLIKE loss function (see Eq. (22)). The forecast produced from the model in the row is compared to the benchmark volatility which is the 5-min realized volatility (high-frequency proxy). Forecast accuracy is compared across all individual and combination forecasts using the model confidence test of Hansen et al. (2011). [a]The indicates that the given model in the row is a part of the superior set of models, suggesting that given models lead to statistically comparable forecasts.

Table 3
Forecast accuracy for five-days-head forecasts (QLIKE).

	WTI	Brent	Natgas	HO
Panel A: High-frequency volatility models				
Standard HAR with RV	0.062[a]	0.048[a]	0.048[a]	0.058[a]
RV-HAR-NV	0.059[a]	0.047[a]	0.047[a]	0.057[a]
RV-HAR-ASY	0.06[a]	0.047[a]	0.047[a]	0.056[a]
RV-ARFIMA-GARCH	0.051[a]	0.044[a]	0.044[a]	0.045[a]
RV-GARCH	0.105	0.053[a]	0.053[a]	0.073
Panel B: Low-frequency volatility models				
Standard HAR with RB	0.084	0.114	0.114	0.108
RB-HAR-NV	0.082	0.115	0.115	0.106
RB-HAR-ASY	0.08	0.113	0.113	0.108
RB-ARFIMA-GARCH	0.087	0.069	0.069	0.128
RB-GARCH	0.105	0.068	0.068	0.079
Panel C: Combining high-frequency models				
Trimmed mean	0.052[a]	0.044[a]	0.044[a]	0.049[a]
Discounted weighted mean	0.082[a]	0.048[a]	0.048[a]	0.055[a]
Average across top three models	0.049[a]	0.043[a]	0.043[a]	0.043[a]
Panel D: Combining low-frequency models				
Trimmed mean	0.072[a]	0.084	0.084	0.087
Discounted weighted mean	0.084[a]	0.068	0.068	0.073
Average across top three models	0.061[a]	0.062	0.062	0.061
Panel E: Combining high- and low-frequency models				
Trimmed mean	0.057[a]	0.052[a]	0.052[a]	0.056[a]
Discounted weighted mean	0.082[a]	0.057[a]	0.057[a]	0.056[a]
Average across top three models	0.051[a]	0.046[a]	0.046[a]	0.046[a]

Note: The values in the table are averages of the QLIKE loss function (see Eq. (22)). The forecast produced from the model in the row is compared to the benchmark volatility which is the 5-min realized volatility (high-frequency proxy). Forecast accuracy is compared across all individual and combination forecasts using the model confidence test of Hansen et al. (2011). [a]The indicates that the given model in the row is a part of the superior set of models, suggesting that given models lead to statistically comparable forecasts.

model relies on the range-based information only in the measurement equation. Nevertheless, individual low-frequency forecasts are vastly inferior to volatility predictions based on high-frequency data for horizons of one to five days (Panels A and B in Tables 2–3).

Table 4
Forecast accuracy for 22-days-head forecasts (QLIKE).

	WTI	Brent	Natgas	HO
Panel A: High-frequency volatility models				
Standard HAR with RV	0.098[a]	0.07[a]	0.095[a]	0.084[a]
RV-HAR-NV	0.095[a]	0.07[a]	0.095[a]	0.082[a]
RV-HAR-ASY	0.096[a]	0.072[a]	0.094[a]	0.081[a]
RV-ARFIMA-GARCH	0.088[a]	0.063[a]	0.094[a]	0.076[a]
RV-GARCH	0.115[a]	0.078[a]	0.101[a]	0.083[a]
Panel B: Low-frequency volatility models				
Standard HAR with RB	0.096[a]	0.121	0.135	0.097[a]
RB-HAR-NV	0.095[a]	0.122	0.136	0.096[a]
RB-HAR-ASY	0.097[a]	0.121	0.134	0.098[a]
RB-ARFIMA-GARCH	0.118	0.085[a]	0.155	0.163
RB-GARCH	0.11[a]	0.071[a]	0.11[a]	0.085[a]
Panel C: Combining high-frequency models				
Trimmed mean	0.087[a]	0.066[a]	0.088[a]	0.073[a]
Discounted weighted mean	0.087[a]	0.068[a]	0.095[a]	0.074[a]
Average across top three models	0.081[a]	0.065[a]	0.083[a]	0.067[a]
Panel D: Combining low-frequency models				
Trimmed mean	0.094[a]	0.094	0.12[a]	0.088[a]
Discounted weighted mean	0.095[a]	0.089	0.126	0.083[a]
Average across top three models	0.082[a]	0.079[a]	0.094[a]	0.08[a]
Panel E: Combining high- and low-frequency models				
Trimmed mean	0.087[a]	0.071[a]	0.095[a]	0.074[a]
Discounted weighted mean	0.096[a]	0.08[a]	0.11[a]	0.078[a]
Average across top three models	0.08[a]	0.068[a]	0.084[a]	0.069[a]

Note: The values in the table are averages of the QLIKE loss function (see Eq. (22)). The forecast produced from the model in the row is compared to the benchmark volatility which is the 5-min realized volatility (high-frequency proxy). Forecast accuracy is compared across all individual and combination forecasts using the model confidence test of Hansen et al. (2011). [a]The indicates that the given model in the row is a part of the superior set of models, suggesting that given models lead to statistically comparable forecasts.

Table 5
Forecast accuracy for 66-days-head forecasts (QLIKE).

	WTI	Brent	Natgas	HO
Panel A: High-frequency volatility models				
Standard HAR with RV	0.187[a]	0.106[a]	0.145[a]	0.166[a]
RV-HAR-NV	0.184[a]	0.105[a]	0.145[a]	0.164[a]
RV-HAR-ASY	0.184[a]	0.107[a]	0.144[a]	0.166[a]
RV-ARFIMA-GARCH	0.187[a]	0.092[a]	0.191[a]	0.19[a]
RV-GARCH	0.179[a]	0.128[a]	0.154[a]	0.134[a]
Panel B: Low-frequency volatility models				
Standard HAR with RB	0.17[a]	0.158	0.17[a]	0.154[a]
RB-HAR-NV	0.167[a]	0.16	0.17[a]	0.152[a]
RB-HAR-ASY	0.167[a]	0.163	0.169[a]	0.154[a]
RB-ARFIMA-GARCH	0.205[a]	0.116[a]	0.287	0.241
RB-GARCH	0.152[a]	0.095[a]	0.152[a]	0.12[a]
Panel C: Combining high-frequency models				
Trimmed mean	0.175[a]	0.101[a]	0.142[a]	0.149[a]
Discounted weighted mean	0.19[a]	0.107[a]	0.206	0.178
Average across top three models	0.176[a]	0.1[a]	0.15[a]	0.156[a]
Panel D: Combining low-frequency models				
Trimmed mean	0.165[a]	0.13	0.163[a]	0.139[a]
Discounted weighted mean	0.169[a]	0.126[a]	0.25	0.152[a]
Average across top three models	0.156[a]	0.118[a]	0.178[a]	0.14[a]
Panel E: Combining high- and low-frequency models				
Trimmed mean	0.165[a]	0.109[a]	0.148[a]	0.139[a]
Discounted weighted mean	0.168[a]	0.114[a]	0.242	0.172[a]
Average across top three models	0.163[a]	0.108[a]	0.163[a]	0.144[a]

Note: The values in the table are averages of the QLIKE loss function (see Eq. (22)). The forecast produced from the model in the row is compared to the benchmark volatility which is the 5-min realized volatility (high-frequency proxy). Forecast accuracy is compared across all individual and combination forecasts using the model confidence test of Hansen et al. (2011). [a]The indicates that the given model in the row is a part of the superior set of models, suggesting that given models lead to statistically comparable forecasts.

For illustrative purposes, we plot the realized variance and selected high-frequency and low-frequency day-ahead volatility model forecasts in Figs. 2–3. It is evident that both forecasts follow the realized variance, but the low-frequency forecasts often appear to have periods

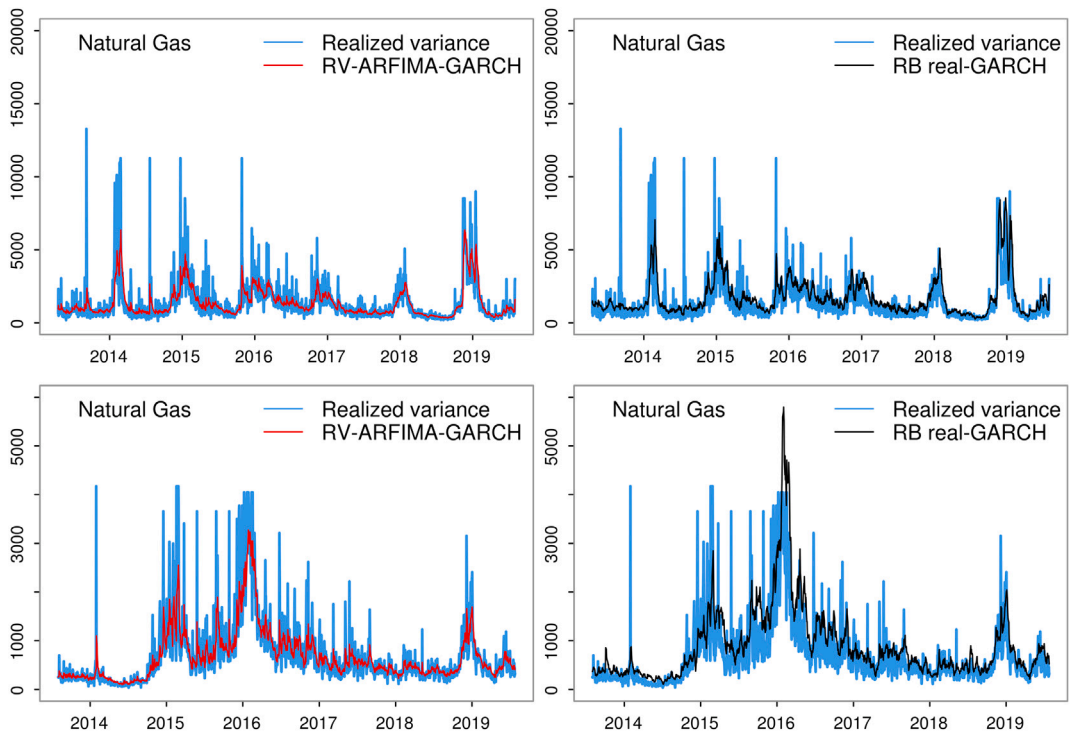

Fig. 2. Observed and predicted realized and range-based volatility: Natural Gas and Heating Oil. *Note:* Blue line is the proxy for the true volatility (5-min realized volatility). The red line corresponds to the series of forecasts generated from the high-frequency volatility model with lowest forecast error (as indicated via QLIKE). The black line corresponds to the series of forecasts generated from the low-frequency volatility model with lowest forecast error (as indicated via QLIKE).

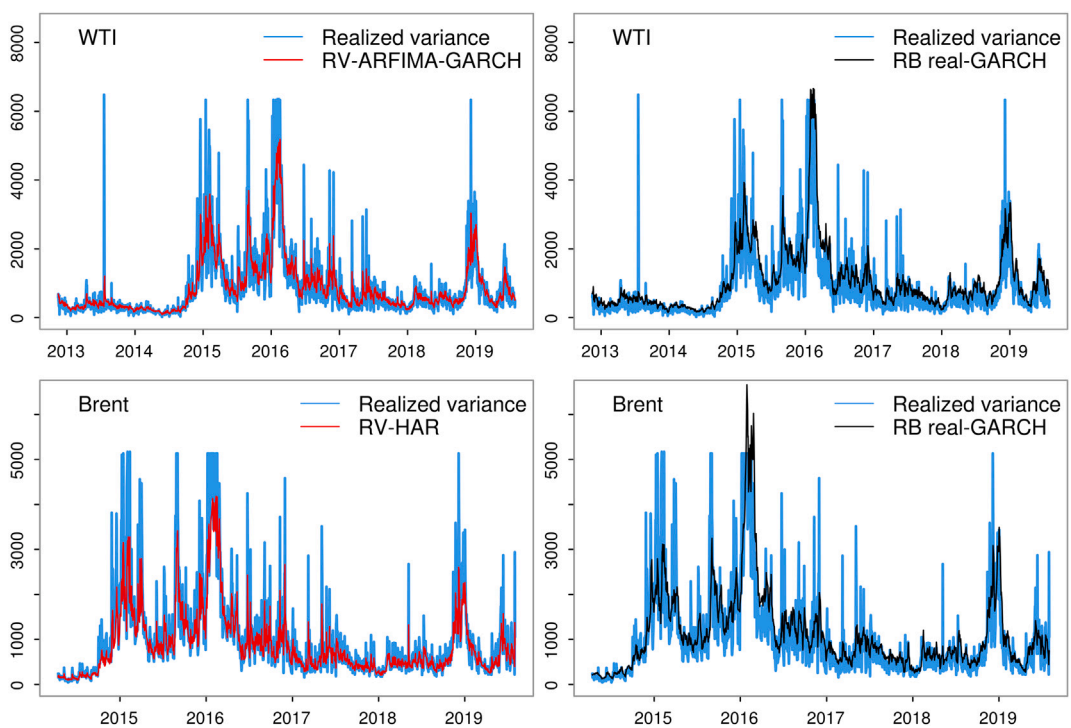

Fig. 3. Observed and predicted realized and range-based volatility: WTI and Brent. *Note:* Blue line is the proxy for the true volatility (5-min realized volatility). The red line corresponds to the series of forecasts generated from the high-frequency volatility model with lowest forecast error (as indicated via QLIKE). The black line corresponds to the series of forecasts generated from the low-frequency volatility model with lowest forecast error (as indicated via QLIKE).

where they systematically overestimate market volatility. This observation suggests that the short-horizon low-frequency volatility forecasts are biased, which is expected, given the descriptive statistics. To give an example of the striking difference between high- and low-frequency models, the RV-HAR-NV model shows an average one-day-ahead QLIKE of 0.028 for Brent, while its corresponding range-based equivalent scores almost ten times worse, with a QLIKE of 0.229. Across all assets, the absolute differences between RV and RB-based five-day ahead

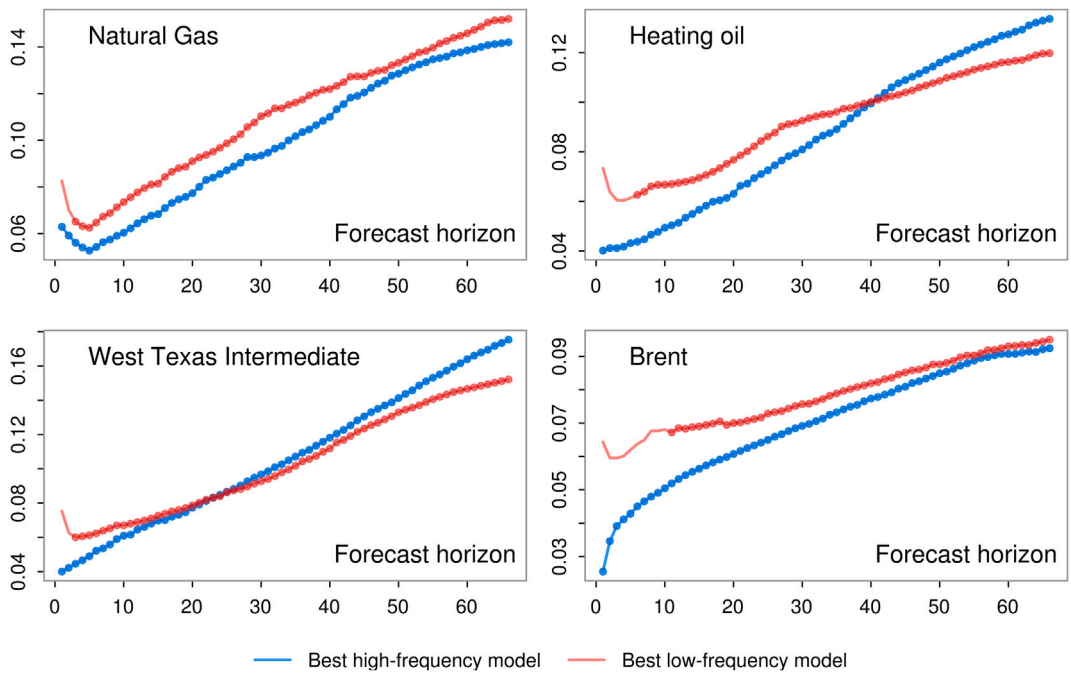

Fig. 4. Predictive performance across forecasting horizons. *Note:* The red (blue) lines denote average QLIKE values of the best-performing low-frequency (high-frequency) volatility model. A dot in the line indicates that the model belongs to the set of superior models as indicated by the model confidence test of Hansen et al. [87].

forecasts decrease considerably compared to one-day ahead predictions, but the forecast errors of RB forecasts are still approximately 1.5 to 2 times larger. Hence, it is not surprising that for these shorter-term forecasting horizons, individual high-frequency volatility models and combinations thereof appear in the superior set of models, as identified by the MCS test, while individual range-based forecasts and their combinations are almost never represented in this set.

The picture uncovered for monthly and quarterly forecasts (Tables 4–5) shows a compelling change. The difference between RV and RB-based performance diminishes abruptly compared to short-term forecasts, making low-frequency volatility models more competitive than their high-frequency counterparts. However, the difference in forecasting accuracy is not statistically significant. In many instances, RB models perform even better in terms of average QLIKE than their high-frequency model forecast counterparts. This phenomenon is particularly pronounced for WTI and heating oil volatility. At these longer-term horizons, individual range-based forecasts are frequently assigned to a superior set of models. Hence, the benefit of using high-frequency data for longer-term forecasts becomes at least challenged by low-frequency models.

A comparison of individual and combination forecasts obtained with both high- and low-frequency data (Panels A-D in Tables 2–5) suggests that combination forecasts are capable of improving the forecasting performance of individual models. This finding is in line with the large body of literature advocating the benefits of combination methods. Combining low-frequency models even at short-term forecasting horizons where individual RB models are strictly inferior to RV models substantially reduces the difference in these performances. In some longer-horizon cases, RB model combinations even yield lower average QLIKE values than combinations of high-frequency models.

In Panels B and D in Tables 2–5, we aimed to test whether low-frequency volatility models are useful when high-frequency data *are not available*. In a recent study, Ma et al. [48] show that in a day-ahead setting, combining high- and low-frequency volatility model forecasts leads to improved accuracy of day-ahead forecasts of stock market indices in the U.S. and China. By combining low- and high-frequency volatility models, in Panel E Tables 2–5, we follow [48] and test whether low-frequency volatility models are useful when high-frequency data *are available*. The results show that for 22- and 66-day ahead forecasts, combinations of RV and RB models lead to QLIKE values that, in several cases, are lower than those generated with high-frequency models only. However, this outcome does not appear to be an empirical rule. Hence, in our setting, it does not appear that low-frequency volatility forecasts contain information incremental to that inherent in high-frequency data-based long-term forecasts.

Finally, Fig. 4 summarizes our key results from this section. The curves capture the performance of best-performing individual or combination realized (in black) and range-based (in red) volatility models predicting volatility 1 to 66 days ahead. The lower the value is, the lower the forecast error. A dot on a curve indicates that the given model belongs to the superior set of models. If both curves have a dot at the same forecast horizon, their performance can be considered statistically similar to each other. The graphs corroborate our findings that high-frequency data-based models perform significantly better in the short term. For longer-term horizons, forecasts requiring solely daily (low-frequency) data exhibit performance that is statistically indistinguishable from that of realized volatility models. This phenomenon occurs on a consistent basis for fortnightly volatility forecasts and for all assets. We also observe that for heating oil and WTI, there is a certain threshold after which low-frequency volatility models tend to outperform high-frequency volatility models.

In the next section, we will investigate how range-based low-frequency volatility estimates compare to high-frequency volatility estimates when applied in a risk management context.

4.3. An application to predict multiple-day-ahead price declines

We present an application of how to apply high- and low-frequency volatility estimates to predict multiple-day-ahead price declines, specifically, the 5% expected shortfall (ES). For that purpose, we use the CSQR-ES model as described in Section 3.6. To improve planning and decision making, the CSQR-ES model can be conveniently used by all stakeholders whose operations are subject to price movements. The existing literature seems to be centered mostly around day-ahead

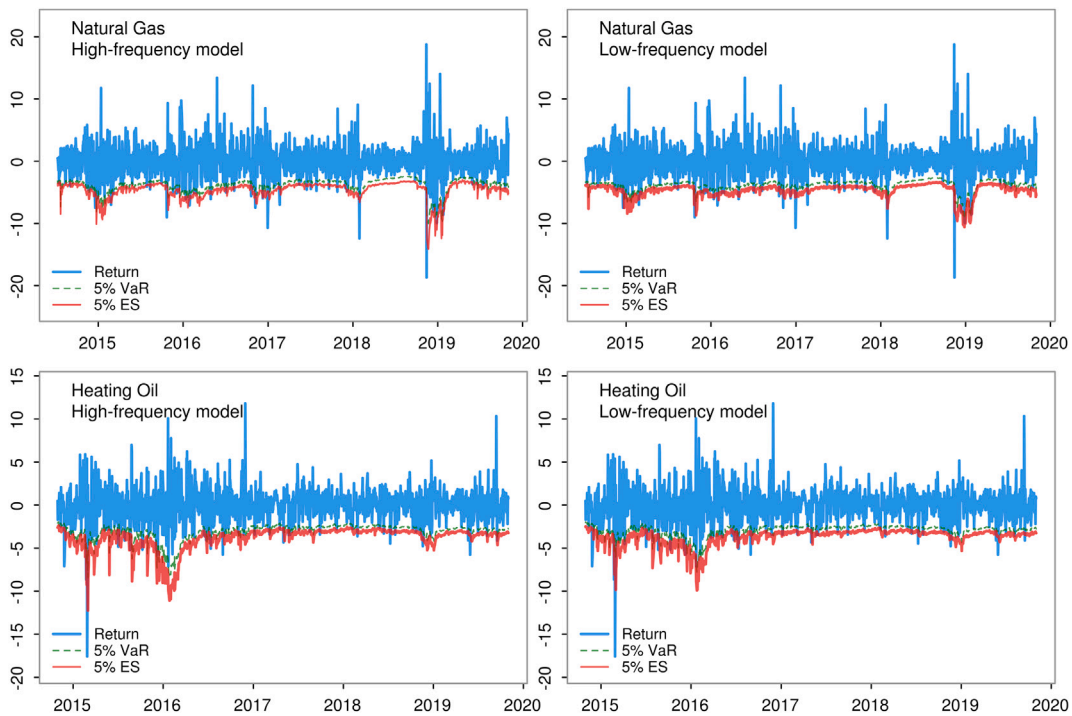

Fig. 5. 5% Expected Shortfall and VaR for Natural Gas and Heating Oil. *Note:* The figure denotes daily close-to-close (R_t) returns, predicted day-ahead 5% VaRs as well as the predicted 5% Expected Shortfall from a selected CSQR-ES model. The comparison shows that low-frequency ES model predictions are more stable over-time.

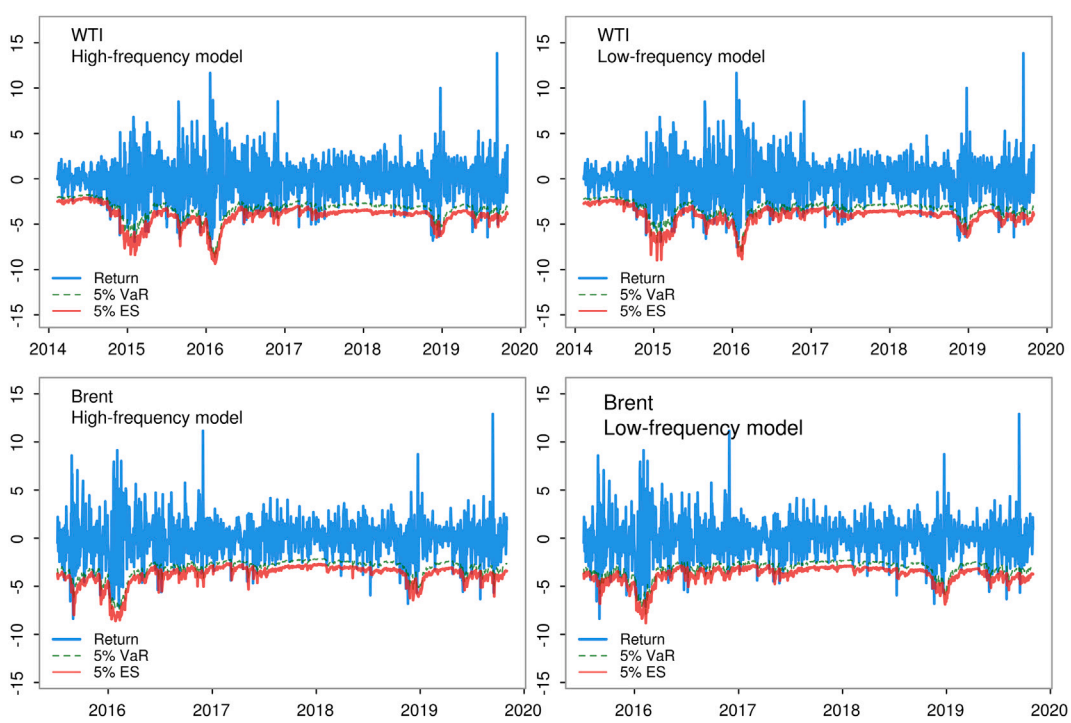

Fig. 6. 5% Expected Shortfall and VaR for WTI and Brent. *Note:* The figure denotes daily close-to-close (R_t) returns, predicted day-ahead 5% VaRs as well as the predicted 5% Expected Shortfall from a selected CSQR-ES model. The comparison shows that low-frequency ES model predictions are more stable over-time.

forecasts of ES (e.g. [99–101]). Given the Basel Committee on Banking Supervision [102] requirements, we instead forecast the ES (risk measure) for next-days'. The goal of this section is twofold. First, we show that predictions from the proposed CSQR-ES model lead to well-specified (accurate) forecasts. Second, we compare the accuracy of ES predictions that use high-frequency or low-frequency volatility estimates. In doing so, we can assess the merit of low-frequency volatility estimated for the purpose of managing market losses.

Our approach to predicting ES is based on summing predicted VaRs at the given quantiles, namely, $5.0\%, 4.0\%, 3.0\%, 2.0\%$ and 1.0%. For illustration purposes, in Figs. 5 and 6, we plot the 5.0% predicted day-ahead VaRs and corresponding 5% ES from the $ES^{h,H}_{\psi=0.9}$ and $ES^{h,L}_{\psi=0.9}$

Table 6
Backtest of day-ahead expected shortfall: CSQR-ES model.

	WTI		Brent		Natgas		Heating oil	
	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$
Panel A: High-frequency expected shortfall model								
Ave	0.287†	1.491†	0.037	1.467†	0.999†	1.612†	0.022	1.434†
$\psi = 0.5$	0.159†	1.491†	0.039	1.468†	0.962†	1.609†	0.065†	1.429†
$\psi = 0.9$	0.283†	1.49†	0.035	1.467†	0.941†	1.611†	0.017	1.431†
$\psi = 1.0$	0.284†	1.491†	0.037	1.467†	0.968†	1.611†	0.028	1.434†
Panel B: Low-frequency expected shortfall model								
Ave	0.652†	1.489†	0.47†	1.463†	0.132†	1.634	0.095†	1.424†
$\psi = 0.5$	0.454†	1.489†	0.301†	1.465†	0.027	1.627†	0.472†	1.419†
$\psi = 0.9$	0.585†	1.488†	0.358†	1.463†	0.252†	1.627†	0.173†	1.421†
$\psi = 1.0$	0.651†	1.489†	0.402†	1.463†	0.145†	1.629†	0.061†	1.425†

Note: In the JT_2 columns, the value denotes the *p*-value of the JT_2 test, and the † symbol denotes a non-rejection of the null hypothesis of the JT_2 test that the model is well-specified. In column $FZ_t^{h,(0)}$, the value is the average value of the $FZ_t^{h,(0)}$ loss function, and the † symbol denotes that the given (row) expected shortfall prediction belongs to the superior set of models, i.e. we cannot statistically distinguish between these forecasts.

Table 7
Backtest of five day-ahead expected shortfall.

	WTI		Brent		Natgas		Heating oil	
	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$
Panel A: High-frequency expected shortfall model								
Ave	0.26†	2.21†	0.51†	2.24	0.91†	2.4	0.04	2.21†
$\psi = 0.5$	0.92†	2.2†	0.76†	2.23	0.37†	2.38†	0.55†	2.2†
$\psi = 0.9$	0.78†	2.23	0.45†	2.23	0.63†	2.43	0.18†	2.23
$\psi = 1.0$	0.29†	2.22†	0.49†	2.21†	0.89†	2.41	0.03	2.22†
Panel B: Low-frequency expected shortfall model								
Ave	0.52†	2.19†	0.95†	2.22†	0.84†	2.37†	0.13†	2.19†
$\psi = 0.5$	0.74†	2.21†	0.97†	2.24	0.25†	2.39	0.23†	2.21†
$\psi = 0.9$	0.72†	2.2†	0.99†	2.2†	0.24†	2.39	0.17†	2.2†
$\psi = 1.0$	0.52†	2.23	0.96†	2.23	0.75†	2.42	0.1†	2.23

Note: In the JT_2 columns, the value denotes the *p*-value of the JT_2 test, and the † symbol denotes a non-rejection of the null hypothesis of the JT_2 test that the model is well-specified. In column $FZ_t^{h,(0)}$, the value is the average value of the $FZ_t^{h,(0)}$ loss function, and the † symbol denotes that the given (row) expected shortfall prediction belongs to the superior set of models, i.e. we cannot statistically distinguish between these forecasts.

Table 8
Backtest of twenty-two day-ahead expected shortfall.

	WTI		Brent		Natgas		Heating oil	
	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$
Panel A: High-frequency expected shortfall model								
Ave	0.3†	3.11	0.02	2.94	0.76†	2.94	0.15†	3.02
$\psi = 0.5$	0.56†	3.04	0.01	2.92	0.05†	2.92†	0.28†	2.99
$\psi = 0.9$	0.5†	3.09	0.01	2.92	0.38†	3.03	0.19†	3.01
$\psi = 1.0$	0.36†	3.03	0.02	2.89†	0.62†	2.99	0.06†	2.97
Panel B: Low-frequency expected shortfall model								
Ave	0.46†	3.01†	0.01	2.9	0.22†	2.89†	0.12†	2.96†
$\psi = 0.5$	0.66†	3.1	0.08†	2.94	0.49†	2.94	0.53†	3.02
$\psi = 0.9$	0.45†	3.01†	0	2.87†	0.36†	2.96	0.2†	2.94†
$\psi = 1.0$	0.38†	3.08	0	2.92	0.52†	3.01	0.12†	3.01

Note: In the JT_2 columns, the value denotes the *p*-value of the JT_2 test, and the † symbol denotes a non-rejection of the null hypothesis of the JT_2 test that the model is well-specified. In column $FZ_t^{h,(0)}$, the value is the average value of the $FZ_t^{h,(0)}$ loss function, and the † symbol denotes that the given (row) expected shortfall prediction belongs to the superior set of models, i.e. we cannot statistically distinguish between these forecasts.

models. The dynamics of the $ES_{\psi=0.9}^{h,H}$ predictions are slightly different from those generated by $ES_{\psi=0.9}^{h,L}$, as during more volatile periods, $ES_{\psi=0.9}^{h,H}$ appears to lead to lower predicted returns, i.e., it seems to be more sensitive to the volatility on the market.

In Tables 6–9, we present the results from the back-testing of the ES. Column JT_2 presents bootstrapped p-values of the JT_2 hypothesis described in Section 3.7. If a given value is highlighted with the † symbol, the hypothesis of interest is not rejected. This means that the prediction of the ES model (in the given row) led to predictions that are well-specified, i.e., that do not show systematic biases. Column $FZ_t^{h,(0)}$ shows the average value of the $FZ_t^{h,(0)}$ loss function, and the † symbol indicates the number of the given model (row) that led to predictions that place it in the superior set of models.

We first observe that for all asset classes, our approach to predicting ES leads to at least one model that is well specified. This holds across all four forecast horizons. However, there are some notable differences across models, assets and forecast horizons. In a day ahead setting, only one high-frequency model, $ES_{\psi=0.5}^{h,H}$ (barely), passed the test for heating oil, while for Brent, none of the high-frequency models passed the test (see Table 6). On the other hand, at least three out of four low-frequency ES models passed the back-testing. In the

Table 9
Backtest of sixty-six day-ahead expected shortfall.

	WTI		Brent		Natgas		Heating oil	
	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$	JT_2	$FZ_t^{h,(0)}$
Panel A: High-frequency expected shortfall model								
Ave	0.84†	3.82	0.45†	3.62	0.77†	3.44	0.59†	3.51
$\psi = 0.5$	0.37†	3.69†	0.65†	3.58	0.68†	3.4†	0.76†	3.4
$\psi = 0.9$	0.58†	3.86	0.6†	3.61	0.5†	3.51	0.55†	3.44
$\psi = 1.0$	0.68†	3.78	0.55†	3.58†	0.84†	3.47	0.69†	3.38
Panel B: Low-frequency expected shortfall model								
Ave	0.26†	3.6†	0.54†	3.57†	0.86†	3.37†	0.57†	3.34†
$\psi = 0.5$	0.38†	3.82	0.6†	3.61	0.95†	3.44	0.55†	3.51
$\psi = 0.9$	0.44†	3.71	0.55†	3.57†	0.88†	3.43	0.4†	3.35†
$\psi = 1.0$	0.39†	3.85	0.66†	3.6	0.75†	3.51	0.47†	3.44

Note: In the JT_2 columns, the value denotes the p-value of the JT_2 test, and the † symbol denotes a non-rejection of the null hypothesis of the JT_2 test that the model is well-specified. In column $FZ_t^{h,(0)}$, the value is the average value of the $FZ_t^{h,(0)}$ loss function, and the † symbol denotes that the given (row) expected shortfall prediction belongs to the superior set of models, i.e. we cannot statistically distinguish between these forecasts.

twenty-two day-ahead setting we found difficulties in passing the back-testing procedure only for Brent (see Table 8). Specifically, only one low-frequency ES model (barely) passed the back-testing procedure. However, in the five- and sixty-six day-ahead settings almost all ES models passed the back-testing procedure. To summarize, across all assets and forecast horizons, we always found at least one low-frequency ES model that passed the back-testing procedure. The same was not true for high-frequency ES models. This outcome leads us to our second key result of this study: low-frequency volatility estimates can be applied to accurately predict multiple day-ahead ES.

We next study the average $FZ_t^{h,(0)}$ losses reported in Panel B of Tables 6–9. In the day-ahead setting we show that almost all models lead to similarly accurate risk forecasts. This finding holds, irrespective of whether the given model passed the back-testing test of Couperier and Leymarie [60]. For five-, twenty-two and sixty-six day-ahead settings the $FZ_t^{h,(0)}$ and the MCS tests leads to smaller sets of superior models, i.e. the $FZ_t^{h,(0)}$ and the MCS tests discriminate between models much better. For all assets and forecast horizons, at least one of the low-frequency ES models is part of the set of superior models. This was not always true for high-frequency ES models. These results show that to arrive at accurate 5% day-ahead ES predictions, one can rely on low-frequency volatility models only.

5. Discussion

Are high-frequency data always needed to generate precise forecasts of risk measures in energy markets? This study attempts to shed light on this question, that is important for practitioners and academia alike. The widely adopted concept of realized volatility is known to yield accurate short-term volatility forecasts for various assets, as supported in theory (e.g., [11,67]) and in empirical studies (e.g., [48,75]). What we have not understood yet is whether low-frequency volatility estimates can help energy market participants manage financial market risks over longer forecast horizons.

As a proxy for low-frequency estimators, we employ daily ranges, which have a long history in finance and are known to be the most efficient low-frequency volatility estimators. These estimators require only daily high, low, open and close prices, i.e., this information is usually publicly available and free of cost as opposed to high-frequency data, which often require a subscription to costly databases. Furthermore, daily price ranges are straightforward to calculate, whereas a market participant who is willing to use realized volatility metrics may face practical issues related to the choice of sampling frequencies and adjustments for microstructure noise arising due to the aggregation of a large number of intraday returns.

The results reported in previous sections show that if high-frequency data are not available, low-frequency volatility estimates can be useful. With medium- and longer-term volatility forecasts, forecasts requiring solely daily data lead to performance that is statistically indistinguishable from that of forecasts from realized volatility models. If volatility forecasts are of interest, our results cannot be interpreted as evidence that high-frequency volatility estimates can be substituted. As in the short-term, high-frequency volatility forecasts are still much more accurate.

Low-frequency volatility estimates appear to be valuable in the context of predicting extreme price declines in energy markets. As high-frequency volatility estimates tend to lead to worse forecasts of extreme price declines, within our modeling framework, we actually recommend using low-frequency volatility estimates to predict extreme price declines.

Results reported above are subject to several limitations. Most prominently, our study is based on the most common high-frequency volatility estimator, the 5 min realized volatility estimators. Therefore other high-frequency estimators might strengthen the case for high-frequency volatility and extreme price forecasts. Finally, an important feature of our study is that we have relied on price data only. We assume that exploring other relevant information (e.g. from other markets, economic news and sentiment) should lead to even smaller difference in volatility and extreme price forecasts. Still, this study has important practical implications. While it is confirmed that realized volatility unquestionably provides the basis for the most accurate short-term volatility forecasts, we demonstrate that low-frequency data still matter and can yield meaningful and precise forecasts at medium- and longer-term horizons and for economic applications (expected shortfall predictions).

6. Conclusion

We compare the performance of daily ranges and realized volatility as applied to a variety of popular volatility forecasting models and combinations thereof. Based on ten years of data on four of the world's most important and liquid energy futures contracts, the West Texas Intermediate crude oil, natural gas and heating oil futures traded at the Chicago Mercantile Exchange and International Commodity Exchange Brent crude oil futures, we first compare volatility forecasting performance at horizons from one to 66 trading days ahead using a high-frequency metric as a benchmark. Our *new finding* is that at forecasting horizons longer than two weeks, forecasts based on daily data only are found to be similarly accurate to realized volatility forecasts. This phenomenon is observed for all considered energy market assets. For longer horizons, combinations of daily range models perform even better in the case of West Texas Intermediate and heating oil than those obtained with high-frequency models only.

We also apply low-frequency volatility estimates to accurately predict extreme price declines in energy markets. In doing so, we make

a methodological contribution by proposing a *novel* complete subset quantile regression model to predict the multiple day-ahead 5% expected shortfall. Our back-testing results and statistical evaluation reveal that this model yields well-specified forecasts for expected shortfalls, particularly when low-frequency volatility estimates are used. The model can be used by all stakeholders in the energy market supply chain (e.g. market participants, producers, governments) whose operations are subject to energy market price movements.

CRediT authorship contribution statement

Štefan Lyócsa: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Funding acquisition, Project administration. **Neda Todorova:** Conceptualization, Methodology, Data curation, Investigation, Writing - original draft, Writing - review & editing. **Tomáš Výrost:** Methodology, Software, Validation, Writing - review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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